

Root Cause Analysis

Expo Android build failure in the mobile app (apps/mobile)

Incident date	2026-09-24
Status	Resolved. Fix in commit 7af4578 ("commit mobile apk issue"), merged to main.
Affected component	Expo (React Native) mobile app, apps/mobile
Severity	Build blocked for the Expo Android app. No production or data impact.

1. Summary

Building the Expo mobile app failed at `npx expo prebuild` with:

```
ENOENT: no such file or directory, open '../apps/mobile/assets/adaptive-icon.png'
```

In the same run, Expo reported *"The android project is malformed"* and deleted the `apps/mobile/android` folder before failing.

2. Impact

- The Expo Android app could not be built.
- **No impact** on the website (task-tracker-web-pi.vercel.app), the native Android app in the root `android/` folder, or any customer data.
- The folder Expo deleted was an exact duplicate of the root `android/` project, so nothing was lost.

3. Root causes

#	Cause	Detail
1	Image files referenced but never committed	<code>apps/mobile/app.json</code> names four images: <code>icon.png</code> , <code>adaptive-icon.png</code> , <code>splash.png</code> and <code>favicon.png</code> . They arrived with the monorepo template migration (a6ed520, 2026-09-23), but the <code>assets/</code> folder was never added. Expo's icon step reads <code>adaptive-icon.png</code> first, so the build stopped there.
2	Hand-written native folder inside the Expo app	The same migration copied the hand-written Android WebView project (root <code>android/</code> , eabd206, 2026-09-19) into <code>apps/mobile/android</code> . Expo generates that folder itself during prebuild, did not recognise the copy, reported it as "malformed" and cleared it on every run.

4. Why it was not caught earlier

- **No automated checks run at all.** The CI workflow files live in `infra/workflows/`, but GitHub only runs workflows from `.github/workflows/`, which does not exist. Even `ci.yml` does not build the mobile app.
- **The mobile scripts cannot fail.** In `apps/mobile/package.json`, *build* only prints "Mobile assets ready", *test* only prints "Mobile tests passed", and *lint* ends in `|| true`, which hides errors.
- **The app had never been built.** Until 2026-09-23 the Expo app was a static mock screen. When it was converted to the WebView app, `app.json` was edited (colours only) without running a prebuild. The hand-over noted the Expo app had not been built or type-checked, so the missing images went unnoticed.

5. Resolution (commit 7af4578)

- Generated the four missing images in the app's brand style (indigo background, white check mark) at the sizes Expo expects.
- Removed the duplicate apps/mobile/android folder and added the Expo-generated android/, ios/ and .expo/ folders to .gitignore (Expo's standard setup).
- Rewrote apps/mobile/README.md with the correct setup and build steps.
- Set the live site address in apps/mobile/config.ts and android/gradle.properties.

Verification: from a clean install, `npx expo prebuild --platform android --clean` completes successfully, launcher icons are generated at every screen density, and the app code type-checks with no errors.

6. Preventive actions (branch feature/tasktracker-06)

#	Action	Status
1	Activate CI: a new, working .github/workflows/ci.yml (the old CI file could not pass and was retired). It runs the web QA suite, an API compile/load check and the mobile checks on every pull request and push to main.	Done
2	Mobile CI job: config and asset check, type-check, expo prebuild and JavaScript bundle. Verified that the check fails on a missing adaptive-icon.png.	Done
3	Mobile build, lint and test scripts now do real checks (no true or placeholder echo commands).	Done
4	Keep a single copy of the native Android WebView project, in the root android/ folder only.	Done
5	Found while adding the checks: the Expo entry point ("main": "App.tsx") never registered the app, so it would have opened to a blank screen. Added index.js with registerRootComponent; the bundle check covers it.	Done

7. Timeline

Date	Event
2026-09-19	Native Android WebView project added in root android/ (eabd206).
2026-09-23	Monorepo template migration (a6ed520): app.json references four images that are not committed; the Android project is duplicated into apps/mobile/android.
2026-09-23	Expo app converted from a static mock to a WebView of the live site; no prebuild run.
2026-09-24	First real expo prebuild: fails with the missing adaptive-icon.png and clears the duplicate Android folder.
2026-09-24	Fix committed (7af4578), verified with a clean prebuild, and merged to main.